Question ID: df32b09c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Tom scored 85, 78, and 98 on his first three exams in history class. Solving which inequality gives the score, G, on Tom's fourth exam that will result in a mean score on all four exams of at least 90 ?

Answer

A. $90-(85+78+98) \leq 4G$

B. $4G+85+78+98 \geq 360$

C. $\frac{(G+85+78+98)}{4} \geq 90$

D. $\frac{(85+78+98)}{4} \geq 90-4G$

Correct Answer: C

Rationale

Choice C is correct. The mean of the four scores (G, 85, 78, and 98) can be expressed as $\frac{G+85+78+98}{4}$. The inequality that expresses the condition that the mean score is at least 90 can therefore be written as $\frac{G+85+78+98}{4} \geq 90$.

Choice A is incorrect. The sum of the scores (G, 85, 78, and 98) isn't divided by 4 to express the mean. Choice B is incorrect and may be the result of an algebraic error when multiplying both sides of the inequality by 4. Choice D is incorrect because it doesn't include G in the mean with the other three scores.